Lecture L03
Graphical Models

What a structured joint distribution assumes, guarantees and makes cheap

Probabilistic Machine Learning 2027-1

YOUR NOTES

Section 1

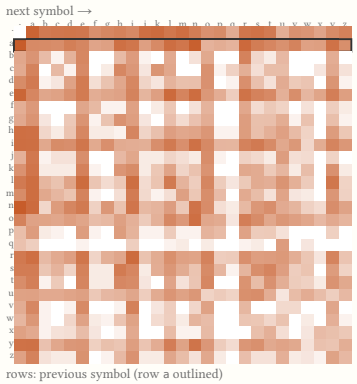
Two models of a name: the full model and L02's simplification

The full model, L02's simplification, and the two assumptions between them

YOUR NOTES

Lecture L03 — Graphical Models

L02 modelled a name with one table of next-character counts



L02 names data; counts recomputed

Row a: what follows an a (33,885 transitions)

next	END	n	r	l	…
count	6,640	5,438	3,264	2,528	
$\hat{P}_{ak}$	0.20	0.16	0.10	0.07	

L02's model and its fit

$p(c_t = k | c_{t-1} = j) = P_{jk}$, fitted as $\hat{P}_{jk} = N_{jk}/N_j$

$p(\text{.emma.}) = \hat{P}_{.,e}\hat{P}_{e,m}\hat{P}_{m,m}\hat{P}_{m,a}\hat{P}_{a,.} \approx 3.5 \times 10^{-6}$

- Sampled from $\hat{P}$ (L02 S7.07): `ssloysthierana`, `imieemmy`, `aucerve`
- Each character sees only the one before it
- Why not condition on more?

1. Name models

S1.02 · Slide 3/56

STUDY NOTE

L02 fitted a model in which each character of a name depends only on the character before it, using one table of next-character frequencies. Strings sampled from it look like names only two letters at a time. The lecture starts by asking what it would cost to condition each character on more of the name.

YOUR NOTES

Lecture L03 — Graphical Models

The full model conditions each character on everything before it

The chain rule: the product rule $p(a, b|u) = p(a|u)p(b|a, u)$, one character at a time

$$p(c_1, ..., c_{L+1}|c_0) = p(c_1|c_0)\, p(c_2|c_0, c_1) \cdots p(c_{L+1}|c_0, ..., c_L)$$

$$= \prod_{t=1}^{L+1} p(c_t|c_0, ..., c_{t-1})$$

`.emma.`	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$
full model	$p(e\|.)$	$p(m\|.e)$	$p(m\|.em)$	$p(a\|.emm)$	$p(.\|.emma)$
L02	$P_{.,e}$	$P_{e,m}$	$P_{m,m}$	$P_{m,a}$	$P_{a,.}$

- No assumption: true for every distribution over names
- L02 keeps only the last character in each factor

1. Name models

S1.03 · Slide 4/56

STUDY NOTE

The chain rule writes the probability of any name as a product of factors, each conditioning one character on everything before it. It is the product rule applied repeatedly, so it holds for every distribution over names and assumes nothing. L02's model shortens every one of these factors to the previous character.

YOUR NOTES

Lecture L03 — Graphical Models

L02's simplification is a conditional-independence assumption

Carried forward S1.03 · Full model (chain rule)

$p(c_1, ..., c_{L+1}|c_0) = p(c_1|c_0)\, p(c_2|c_0, c_1) \cdots p(c_{L+1}|c_0, ..., c_L) = \prod_{t=1}^{L+1} p(c_t|c_0, ..., c_{t-1})$

What L02 assumed: each character depends only on the one before it

$p(c_t|c_0, ..., c_{t-1}) = p(c_t|c_{t-1}), \quad \text{e.g. } p(a|.emm) = p(a|m)$

$$p(c_1, ..., c_{L+1}|c_0) = p(c_1|c_0)\, p(c_2|c_1) \cdots p(c_{L+1}|c_L) = \prod_{t=1}^{L+1} p(c_t|c_{t-1})$$

What conditional independence means

$X \perp Y \mid U$ means $p(x|y, u) = p(x|u)$: once U is known, Y adds nothing about X.

Here $C_t \perp (C_0, ..., C_{t-2}) \mid C_{t-1}$ when C_t is the tth character.

- The chain rule is true for all random variables; conditional independence is a specific claim.

1. Name models

S1.04 · Slide 5/56

STUDY NOTE

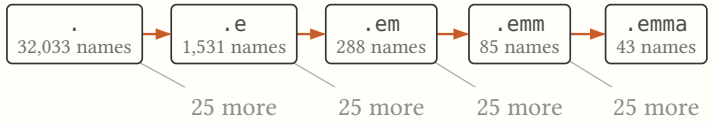
Shortening each factor to the previous character is an assumption: given the previous character, the current one does not depend on the earlier ones. This is conditional independence. Unlike the chain rule it can be false for real names, and it is the first of the two simplifications behind L02's model.

YOUR NOTES

Lecture L03 — Graphical Models

Each model is a table with one row per context

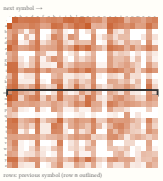
Full model: one row for every prefix — following `.emma` down the prefix tree



The row for `.emm` (what follows "emm") is estimated from the 85 names that start that way.

L02: one row for each previous character

27 rows: "." and a, …, z, whatever the position. The row for m pools every m in every name: 6,642 transitions.



- A *context* is whatever a factor conditions on: each context gets its own row.
  - Full model: the whole prefix, so the contexts multiply by 26 with each character.
  - L02: the previous character only, so 27 contexts at every position.

STUDY NOTE

A factor's context is whatever it conditions on, and each context needs its own row of probabilities. In the full model the context is the whole prefix, so the number of rows grows by a factor of 26 with every character. L02 uses only the previous character, so the same 27 rows serve every position.

YOUR NOTES

Lecture L03 — Graphical Models

Counting rows: millions for the full model, 27 for L02

Full model: one row for each prefix that could precede character t (26 free probabilities per row)

character t	rows needed: one per possible prefix, 26^{t-1}	prefixes that occur in the 32,033 names	names per occurring prefix (median)
2	26	26	1,038
3	676	351	21
4	17,576	2,836	4
5	456,976	9,265	2
6	11,881,376	15,448	1

Row $t = 4$: the fourth character needs a row for each of 17,576 three-letter prefixes; only 2,836 occur, the typical one in 4 names.

L02

27 rows × 26 free probabilities = 702, estimated from 228,146 transitions: about 325 per probability.

- By the sixth character, almost every row has one name or none
- Estimability, not elegance, forced the simplification

STUDY NOTE

Counting rows shows why L02 had to simplify. The full model needs a row for every possible prefix, millions by the sixth character, and almost all of them occur in one name or in none, so they cannot be estimated. Each of L02's 702 probabilities is estimated from about 325 transitions.

YOUR NOTES

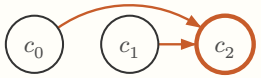
Lecture L03 — Graphical Models

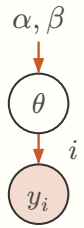
A graph draws what each factor conditions on

A rule for drawing a model

One node per variable. For each factor $p(x|\ldots)$, draw an arrow into x from every variable it conditions on.

The factor $p(c_2|c_0, c_1)$ gives two arrows into c_2:





L01: "arrows specify the modeled conditional relationships"

- L01's coin: $\theta \to y_i$ because $p(y_i|\theta)$ conditions on θ
- A missing arrow: the factor ignores that variable

1. Name models

S1.07 · Slide 8/56

STUDY NOTE

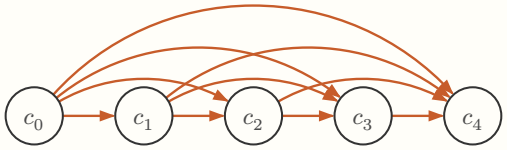
A graphical model draws one node per variable and an arrow into each variable from every variable its factor conditions on. L01's coin model followed this rule: each toss conditions on the coin's bias, so an arrow runs from the bias to every toss. A missing arrow means that a factor ignores that variable.

YOUR NOTES

Lecture L03 — Graphical Models

The full model's graph has every arrow; L02's keeps one per character

Full model: c_t conditions on every earlier character



L02: c_t conditions on c_{t-1} only



- Each removed arrow is a conditional-independence assumption
- Into c_4: four arrows become one

1. Name models

S1.08 · Slide 9/56

STUDY NOTE

Drawn by this rule, the full model has an arrow from every earlier character into each character, while L02's model keeps only the arrow from the previous character. Each removed arrow is one conditional-independence assumption, so the graph shows directly what the simplification discarded.

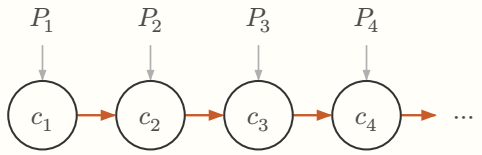
YOUR NOTES

Lecture L03 — Graphical Models

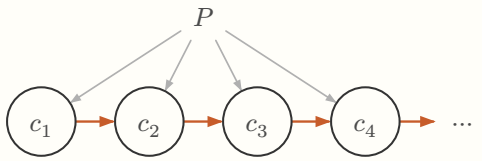
L02 made two simplifications, not one

From the full model to L02 in two steps

model	factor for c_t	free probabilities
full	$p(c_t \mid c_0, \ldots, c_{t-1})$	$26 \cdot 26^{t-1}$ at position t
and conditional independence	$p(c_t \mid c_{t-1}) = P^{(t)}_{c_{t-1} c_t}$	16 positions × 702
and parameter tying	$P^{(t)} = P$ for every t	one table: 702



conditional independence alone: a table $P^{(t)}$ for each position



and parameter tying: one table P for every position

- Conditional independence: shorter contexts
- Parameter tying: every position uses the same table, so counts pool
- Arrows show the first; the shared node P shows the second

1. Name models

S1.09 · Slide 10/56

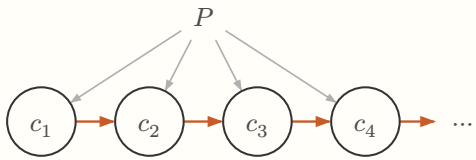
STUDY NOTE

L02 made two separate assumptions. Conditional independence shortens each context to the previous character. Parameter tying uses the same table at every position, so the counts from all positions pool into one table. Either could hold without the other. Arrows show the first; a single shared node for the table shows the second.

YOUR NOTES

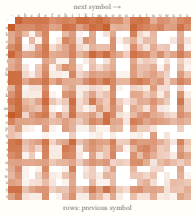
Lecture L03 — Graphical Models

Section 1 summary: Two simplifications made names estimable



each character depends only on the one before; one table P for all positions

$P =$



TAKE HOME

- The full model needs a row for every prefix: millions, mostly without data.
- Keeping only the previous character and one shared table leaves 702 probabilities — and the graph shows what was kept.

STUDY NOTE

The full model of a name cannot be estimated from the data available. L02's model is estimable because of two assumptions: conditional independence, which shortens every context to the previous character, and parameter tying, which shares one table across positions. A graph records the first through the arrows it leaves out.

YOUR NOTES

Section 2

Build a model one variable at a time: the student record

Choosing, one variable at a time, what each factor conditions on

YOUR NOTES

Lecture L03 — Graphical Models

A general problem: simplifying multivariate joint distributions

A name (Section 1)

`.emma.`

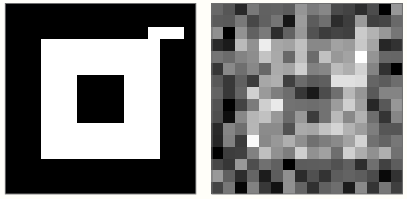
$c_0 c_1 \cdots c_5 =$ `.emma.`

each c_t: one of 27 symbols

Modeling student performance

I intelligence	Low / **High**
D difficulty	**Easy** / Hard
G grade	**A** / B / C
S SAT	Bad / **Good**
L letter	Bad / **Good**

A noisy binary scan



clean observed

Course-generated image

- Simplification through conditional independence
- Simplification through parameter tying
  - several distributions share one set of parameters

2. Directed graphs

S2.02 · Slide 13/56

STUDY NOTE

Names, student records and images are all joint distributions over many variables, and the full joint distribution is too large to specify or estimate. Conditional independence and parameter tying are general tools for making such distributions manageable. The rest of the lecture applies them to a student record and to an image.

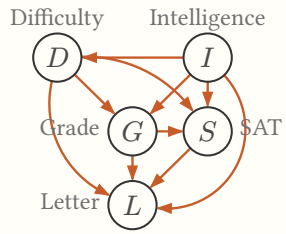
YOUR NOTES

Lecture L03 — Graphical Models

The full model for student performance requires 47 free probabilities

The chain rule in the order I, D, G, S, L

$$p(i, d, g, s, l) = p(i)p(d|i)p(g|i, d)p(s|i, d, g)p(l|i, d, g, s)$$



Every arrow over the variables of Murphy B2 Fig. 4.2

- Simplify one variable at a time
  - what must each factor condition on?

STUDY NOTE

The student record has five variables with 48 possible combinations of values, so a general distribution over records needs 47 free probabilities. The chain rule in the order intelligence, difficulty, grade, SAT, letter writes it as five factors, and its graph has every possible arrow. Next we decide what each factor needs to condition on.

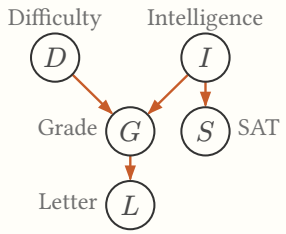
YOUR NOTES

Lecture L03 — Graphical Models

Reduce the full model one assumption at a time

The chain rule in the order I, D, G, S, L, then one assumption per step

$$\begin{aligned} p(i,d,g,s,l) &= p(i)\,p(d|i)\,p(g|i,d)\,p(s|i,d,g)\,p(l|i,d,g,s) \\ &= p(i)\,p(d)\,p(g|i,d)\,p(s|i,d,g)\,p(l|i,d,g,s) \\ &= p(i)\,p(d)\,p(g|i,d)\,p(s|i)\,p(l|i,d,g,s) \\ &= p(i)\,p(d)\,p(g|i,d)\,p(s|i)\,p(l|g) \end{aligned}$$



Ends at Murphy B2 §4.2.2.2, Fig. 4.2

- Each step makes one assumption, which erases arrows and shortens one factor
- $D \perp I$: a class's difficulty says nothing about a student's intelligence
- $S \perp \{D, G\} \mid I$: the SAT reflects ability, not this class or its grade
- $L \perp \{I, D, S\} \mid G$: the letter is written from the grade alone

STUDY NOTE

Each step makes one conditional-independence assumption about a single variable, erases the arrows it drops and shortens that variable's factor. Difficulty is assumed independent of intelligence, the SAT depends only on intelligence, and the letter only on the grade. The grade keeps both parents. The result is the student network of Murphy's Figure 4.2.

YOUR NOTES

Lecture L03 — Graphical Models

One table per variable, one row per value of its parents

The reduced model, filled in one table at a time

$p(I)$

Low	High
0.70	0.30

$p(D)$

Easy	Hard
0.60	0.40

$p(G|I, D)$

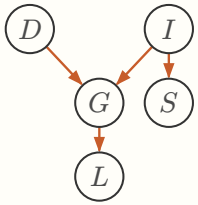
I	D	A	B	C
Low	Easy	0.30	0.40	0.30
Low	Hard	0.05	0.25	0.70
High	Easy	0.90	0.08	0.02
High	Hard	0.50	0.30	0.20

$p(S|I)$

I	Bad	Good
Low	0.95	0.05
High	0.20	0.80

$p(L|G)$

G	Bad	Good
A	0.10	0.90
B	0.40	0.60
C	0.99	0.01



Murphy B2 Fig. 4.2

- Murphy's illustrative numbers (Book 2 Fig. 4.2), not fitted to data

STUDY NOTE

Each variable now has one table, with one row for each combination of its parents' values. The numbers are Murphy's illustrative values, not fitted to data. Fixing the parents selects a row, as the previous character selected a row of L02's table; with data, each row would be estimated from the records with that parent configuration.

YOUR NOTES

Lecture L03 — Graphical Models

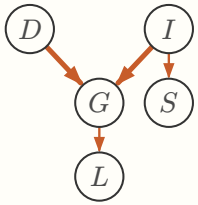
A variable's parents determine its factor in the joint

Parents

- The *parents* of X_i, written $\mathrm{pa}(i)$: the variables with an arrow into X_i
- X_i is their *child*
- Student record: $\mathrm{pa}(G) = \{I, D\}$; $\mathrm{pa}(I) = \varnothing$
- Names: $\mathrm{pa}(C_t) = \{C_{t-1}\}$

The joint is the product of each variable's distribution given its parents:

$$p(i,d,g,s,l) = p(i)\,p(d)\,p(g|i,d)\,p(s|i)\,p(l|g)$$
$$= \prod_k p\left(x_k | x_{\mathrm{pa}(k)}\right)$$



Murphy B2 Fig. 4.2

- For discrete variables with discrete parents:
  - one table per variable: a *conditional probability distribution* (CPD)
  - one row per combination of parent values
- Arrows never form a cycle so we have a *directed acyclic graph* (DAG)

2. Directed graphs

S2.06 · Slide 17/56

STUDY NOTE

A variable's parents are the variables with arrows into it, and the joint distribution is the product of each variable's distribution given its parents. Because every arrow points from an earlier variable to a later one, the graph has no cycles: it is a directed acyclic graph. The arrows record what each factor conditions on, not causation.

YOUR NOTES

Lecture L03 — Graphical Models

One complete record selects one entry from each of the five tables

Selected row (parent values) and entry (own value)

$p(I)$

Low	High
0.70	**0.30**

$p(D)$

Easy	Hard
0.60	0.40

$p(G|I,D)$

I	D	A	B	C
Low	Easy	0.30	0.40	0.30
Low	Hard	0.05	0.25	0.70
High	Easy	**0.90**	0.08	0.02
High	Hard	0.50	0.30	0.20

$p(S|I)$

I	Bad	Good
Low	0.95	0.05
High	0.20	**0.80**

$p(L|G)$

G	Bad	Good
A	0.10	**0.90**
B	0.40	0.60
C	0.99	0.01

Each value under its factor

$$p(\text{High}, \text{Easy}, A, \text{Good}, \text{Good}) =$$

$$\underset{0.30}{p(I = \text{High})} \cdot \underset{0.60}{p(D = \text{Easy})} \cdot \underset{0.90}{p(G = A|\text{High}, \text{Easy})} \cdot \underset{0.80}{p(S = \text{Good}|\text{High})} \cdot \underset{0.90}{p(L = \text{Good}|A)} = 0.11664$$

- A factor’s conditioning values name its row
- The node’s own value names the entry
- A record costs five lookups and one product

2. Directed graphs

S2.07 · Slide 18/56

STUDY NOTE

The probability of one complete record is found by lookup. Each factor's conditioning values pick a row of its table, the variable's own value picks an entry, and the five entries are multiplied. Every later probability in the lecture is built from this calculation.

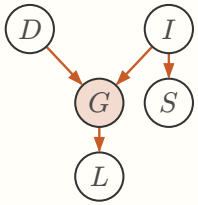
YOUR NOTES

Lecture L03 — Graphical Models

Given a grade of C: sum the records that match, then divide

Every (i, d) with $G = C$; the SAT and the letter are not observed

i	d	$p(i)$	$p(d)$	$p(C\|i,d)$	product
Low	Easy	0.70	0.60	0.30	0.126
Low	Hard	0.70	0.40	0.70	0.196
High	Easy	0.30	0.60	0.02	0.0036
High	Hard	0.30	0.40	0.20	0.024



Murphy B2 Fig. 4.2

Marginal: $P(G = C) = \sum_{i,d} p(i)\, p(d)\, p(C|i, d) = 0.350$

Conditional: $P(I = \text{High}|C) = (0.0036 + 0.024)/0.350 = 0.079$

- Each row is a product of lookups, as for one record (S2.07)
- A marginal sums over what is not observed
  - the SAT and letter tables sum to one, so they drop out
- A conditional divides the matching rows' sum by the total. Murphy reports 0.08 (Book 2 §4.2.6.1)

STUDY NOTE

When some variables are not observed, a probability is the sum of the full products over their values. Here the SAT and letter tables sum to one and drop out, leaving four rows. Dividing the sum of the rows that meet a further condition by the total gives a conditional probability: high intelligence given a C has probability 0.079.

YOUR NOTES

Lecture L03 — Graphical Models

Counting the model's free probabilities: 15 instead of 47

Each row sums to one, so it has one free probability fewer than its entries

$p(I)$

Low	High
0.70	0.30

1 row × 1 = 1

$p(D)$

Easy	Hard
0.60	0.40

1 row × 1 = 1

$p(G|I, D)$

I	D	A	B	C
Low	Easy	0.30	0.40	0.30
Low	Hard	0.05	0.25	0.70
High	Easy	0.90	0.08	0.02
High	Hard	0.50	0.30	0.20

4 rows × 2 = 8

$p(S|I)$

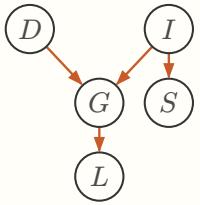
I	Bad	Good
Low	0.95	0.05
High	0.20	0.80

2 rows × 1 = 2

$p(L|G)$

G	Bad	Good
A	0.10	0.90
B	0.40	0.60
C	0.99	0.01

3 rows × 1 = 3



Murphy B2 Fig. 4.2

$1 + 1 + 8 + 2 + 3 = 15$ free probabilities. One table over all $2 \cdot 2 \cdot 3 \cdot 2 \cdot 2 = 48$ possible records would need 47.

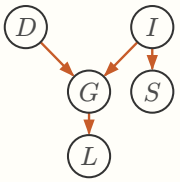
STUDY NOTE

Each row of a table sums to one, so it has one free probability fewer than it has entries. Adding up the five tables gives 15 free probabilities, against 47 for a single table over all records. The conditional-independence assumptions are what keep each table small.

YOUR NOTES

Lecture L03 — Graphical Models

Section 2 summary: A DAG keeps each variable's parents



$p(I)$

Low	High
0.70	0.30

$p(D)$

Easy	Hard
0.60	0.40

$p(G \mid I, D)$

I	D	A	B	C
Low	Easy	0.30	0.40	0.30
Low	Hard	0.05	0.25	0.70
High	Easy	0.90	0.08	0.02
High	Hard	0.50	0.30	0.20

$p(S \mid I)$

I	Bad	Good
Low	0.95	0.05
High	0.20	0.80

$p(L \mid G)$

G	Bad	Good
A	0.10	0.90
B	0.40	0.60
C	0.99	0.01

TAKE HOME

- Each variable keeps only its parents.
- One table per variable, one row per parent combination.
- 15 numbers instead of 47.

2. Directed graphs

S2.10 · Slide 21/56

STUDY NOTE

Section 2 built a model by deciding, one variable at a time, which earlier variables its factor needs. The result is a directed acyclic graph with one table per variable and one row per combination of parent values. The student network needs 15 numbers instead of 47, and probabilities are computed from it by lookups, sums and division.

YOUR NOTES

Section 3

Three-node patterns: what every product over a graph guarantees

When conditioning removes a dependence and when it creates one

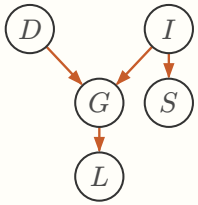
YOUR NOTES

Lecture L03 — Graphical Models

How does adding information affect conditional probabilities?

Three questions about the student record

- Does a good SAT score make an A more likely?
  - What if intelligence is known?
- Does intelligence change what we expect to find in the letter?
  - What if we know the grade as well?
- Are intelligence and difficulty related?
  - What if we know the grade as well?



Murphy B2 Fig. 4.2

The model

$$p(i, d, g, s, l) = p(i)\, p(d)\, p(g|i, d)\, p(s|i)\, p(l|g)$$

- Independences follow from the product, for any tables
- Dependences hold for these tables, and generally

STUDY NOTE

Once a model is a product of small tables, we can ask how learning one variable changes what we expect of another. The three questions involve three different arrangements of three variables. Independences that follow from the product hold for every choice of tables; the dependences hold for these tables and for almost every other choice.

YOUR NOTES

Lecture L03 — Graphical Models

A fork: variables sharing a parent are dependent until it is known

Carried forward S2.05 · Tables and a derived table

$p(I)$

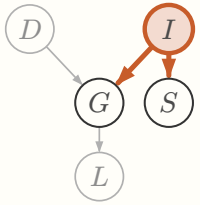
Low	High
0.70	0.30

$p(S|I)$

I	Bad	Good
Low	0.95	0.05
High	0.20	0.80

$p(G = A|I)$, difficulty summed out

I	$G = A$
Low	0.20
High	0.74



Murphy B2 Fig. 4.2

Intelligence not observed

$P(G = A, S = \text{Good}) = \sum_i p(i)\, p(A|i)\, p(\text{Good}|i) = 0.185$

$P(G = A) \cdot P(S = \text{Good}) = 0.362 \cdot 0.275 = 0.100$ **not equal: dependent**

Intelligence observed: $I = \text{High}$

$P(G = A, S = \text{Good}|\text{High}) = p(A|\text{High}) \cdot p(\text{Good}|\text{High}) = 0.74 \cdot 0.80 = 0.592$

The joint is the product of the two conditionals, for either value of I **independent**

- Fork: the shared parent carries the dependence
  - observing it blocks it

3. Three-node patterns

S3.03 · Slide 24/56

STUDY NOTE

The grade and the SAT share a parent, intelligence. Without knowing intelligence they are dependent: a good grade is evidence of high intelligence, which makes a good SAT more likely. Once intelligence is known, the product splits into a function of the grade times a function of the SAT, and they become independent.

YOUR NOTES

Lecture L03 — Graphical Models

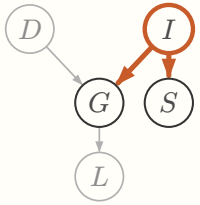
Fork calculation: sum out difficulty, then sum over intelligence

Difficulty summed out: $p(A|i) = \sum_d p(d)\, p(A|i,d)$

i	$p(\text{Easy})\, p(A|i,\text{Easy}) + p(\text{Hard})\, p(A|i,\text{Hard})$	$p(A|i)$
Low	$0.60 \cdot 0.30 + 0.40 \cdot 0.05$	0.20
High	$0.60 \cdot 0.90 + 0.40 \cdot 0.50$	0.74

Sum over intelligence; the letter's table sums to one and drops out

i	$p(i)$	$p(A|i)$	$p(\text{Good}|i)$	$p(i)\, p(A|i)$	$p(i)\, p(\text{Good}|i)$	$p(i)\, p(A|i)\, p(\text{Good}|i)$
Low	0.70	0.20	0.05	0.140	0.035	0.0070
High	0.30	0.74	0.80	0.222	0.240	0.1776
sum				0.362	0.275	0.1846



Murphy B2 Fig. 4.2

Given $I = \text{High}$: $P(A, \text{Good}|\text{High}) = 0.1776/0.30 = 0.592 = 0.74 \cdot 0.80$

- Summing out difficulty weights each of its values and adds (S2.08)
- Column sums: $P(A) = 0.362$, $P(\text{Good}) = 0.275$, $P(A, \text{Good}) = 0.1846 \neq 0.362 \cdot 0.275 = 0.100$
- Fixing I keeps one row
  - divided by $p(\text{High})$, the joint is the product of two lookups

3. Three-node patterns

S3.04 · Slide 25/56

STUDY NOTE

This slide shows where the fork's numbers come from. Difficulty is summed out first, giving the probability of an A for each level of intelligence. Summing over intelligence then gives each marginal and the joint probability as column sums. Fixing intelligence keeps a single row, in which the joint is the product of two lookups.

YOUR NOTES

Lecture L03 — Graphical Models

A pipe: influence flows through the middle until the middle is known

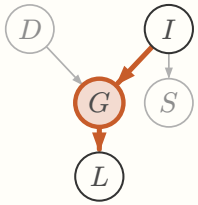
Carried forward S2.05 · Tables and a derived table

$p(G|I)$, difficulty summed out

I	A	B	C
Low	0.20	0.34	0.46
High	0.74	0.17	0.09

$p(L|G)$

G	Bad	Good
A	0.10	0.90
B	0.40	0.60
C	0.99	0.01



Murphy B2 Fig. 4.2

Grade not observed

$P(I = \text{High}, L = \text{Good}) = p(\text{High}) \sum_g p(g|\text{High})\, p(\text{Good}|g) = 0.230$

$P(I = \text{High}) \cdot P(L = \text{Good}) = 0.30 \cdot 0.502 = 0.151$ **not equal: dependent**

Grade observed: $G = A$

$P(I = \text{High}, L = \text{Good}|A) = P(I = \text{High}|A) \cdot p(\text{Good}|A) = 0.613 \cdot 0.90 = 0.552$

The letter's table looks only at the grade, so the joint is the product **independent**

- Pipe: dependence flows through the middle
  - observing the middle blocks it

3. Three-node patterns

S3.05 · Slide 26/56

STUDY NOTE

Intelligence affects the letter only through the grade. Without the grade, a student's intelligence changes the chance of a good letter. Once the grade is known, the letter's table depends only on the grade, so intelligence tells us nothing more about the letter. This is the assumption L02 made about consecutive characters.

YOUR NOTES

Lecture L03 — Graphical Models

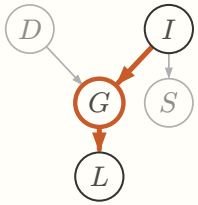
Pipe calculation: the grade's probabilities, then sum over the grade

Difficulty summed out, as for the fork

g	$p(g\|\text{Low})$	$p(g\|\text{High})$	$P(g)$	$p(\text{Good}\|g)$
A	0.20	0.74	0.362	0.90
B	0.34	0.168	0.288	0.60
C	0.46	0.092	0.350	0.01

Sum over the grade

g	$P(g)\,p(\text{Good}\|g)$	$p(g\|\text{High})\,p(\text{Good}\|g)$
A	0.326	0.666
B	0.173	0.101
C	0.003	0.001
sum	0.502	0.768



Murphy B2 Fig. 4.2

Joint: $P(\text{High}, \text{Good}) = 0.30 \cdot 0.768 = 0.230 \neq 0.30 \cdot 0.502 = 0.151$

Given $G = A$: $P(\text{High}|A) = 0.30 \cdot 0.74/0.362 = 0.613$

$P(\text{High}, \text{Good}|A) = 0.613 \cdot 0.90 = 0.552 = P(\text{High}|A) \cdot p(\text{Good}|A)$

- $P(g) = 0.70\,p(g|\text{Low}) + 0.30\,p(g|\text{High})$: intelligence summed out
- Column sums give $P(\text{Good}) = 0.502$ and $p(\text{Good}|\text{High}) = 0.768$
  - both sum over the unobserved grade
- Given A, the letter's row is fixed, so the joint is the product
  - $P(\text{High}|A)$ is High's share of $P(A)$, as in S2.08

3. Three-node patterns

S3.06 · Slide 27/56

STUDY NOTE

The pipe's numbers come from summing over the unobserved grade. Each grade's probability is found by summing out difficulty and then intelligence, and the probability of a good letter is a weighted sum over the three grades. Given an A, the letter's row is fixed and the joint probability becomes a product.

YOUR NOTES

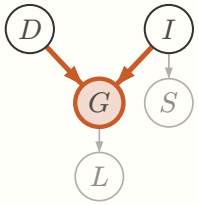
Lecture L03 — Graphical Models

A collider: two parents of one child are independent until it is known

Carried forward S2.08 · Records with $G = C$

i	d	$p(i)$	$p(d)$	$p(C\|i,d)$	product
Low	Easy	0.70	0.60	0.30	0.126
Low	Hard	0.70	0.40	0.70	0.196
High	Easy	0.30	0.60	0.02	0.0036
High	Hard	0.30	0.40	0.20	0.024

$P(C) = 0.350, \quad P(I = \text{High}|C) = 0.079, \quad P(D = \text{Hard}|C) = 0.629$



Murphy B2 Fig. 4.2

Grade not observed

$P(I = \text{High}, D = \text{Hard}) = 0.30 \cdot 0.40 = 0.12$
$= P(I = \text{High}) \cdot P(D = \text{Hard})$ **independent**

Grade observed: $G = C$

$P(I = \text{High}, D = \text{Hard}|C) = 0.024/0.350 = 0.069$, the (High, Hard) row
$P(I = \text{High}|C) \cdot P(D = \text{Hard}|C) = 0.079 \cdot 0.629 = 0.050$ **not equal: dependent**

- Collider: observing the child couples the parents, as a hard class explains away a C

3. Three-node patterns

S3.07 · Slide 28/56

STUDY NOTE

Intelligence and difficulty are independent parents of the grade: before the grade is known, summing over it removes its table. Once a C is observed they become dependent. A hard class is one explanation of the C, so learning that the class was hard makes high intelligence more likely. This is called explaining away.

YOUR NOTES

Lecture L03 — Graphical Models

A collider's descendant: observing it also couples the parents

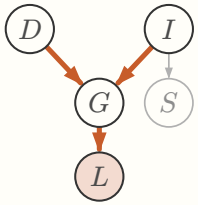
Carried forward S2.05 · Tables

$p(G|I, D)$

I	D	A	B	C
Low	Easy	0.30	0.40	0.30
Low	Hard	0.05	0.25	0.70
High	Easy	0.90	0.08	0.02
High	Hard	0.50	0.30	0.20

$p(L|G)$

G	Bad	Good
A	0.10	0.90
B	0.40	0.60
C	0.99	0.01



Murphy B2 Fig. 4.2

Nothing observed

As for the collider: $P(I = \text{High}, D = \text{Hard}) = 0.12 = P(I = \text{High}) \cdot P(D = \text{Hard})$ **independent**

Letter observed: $L = \text{Bad}$, grade unknown

$p(i, d|\text{Bad}) \propto p(i)\, p(d) \sum_g p(g|i, d)\, p(\text{Bad}|g)$: the grade's table still links i and d

$P(I = \text{High}, D = \text{Hard}|\text{Bad}) = 0.089$

$P(I = \text{High}|\text{Bad}) \cdot P(D = \text{Hard}|\text{Bad}) = 0.140 \cdot 0.538 = 0.075$ **dependent**

- Observing a collider's descendant also opens it, as the letter is evidence about the grade

3. Three-node patterns

S3.08 · Slide 29/56

STUDY NOTE

Observing the letter instead of the grade also makes intelligence and difficulty dependent, because a bad letter is evidence about the grade. The letter weights each grade by how well it predicts the letter, and that weighted sum depends on intelligence and difficulty together. Observing any descendant of a collider opens it.

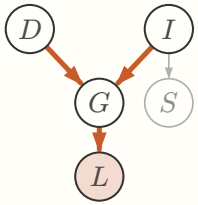
YOUR NOTES

Lecture L03 — Graphical Models

Descendant calculation: the letter weighs each (i, d) record

Every (i, d) with $L = \text{Bad}$; the grade is not observed, so it is summed out

i	d	$p(i)$	$p(d)$	$w(i,d) = \sum_g p(g\|i,d)\,p(\text{Bad}\|g)$	product
Low	Easy	0.70	0.60	0.487	0.2045
Low	Hard	0.70	0.40	0.798	0.2234
High	Easy	0.30	0.60	0.142	0.0255
High	Hard	0.30	0.40	0.368	0.0442



Murphy B2 Fig. 4.2

e.g. $w(\text{Low}, \text{Easy}) = 0.30 \cdot 0.10 + 0.40 \cdot 0.40 + 0.30 \cdot 0.99 = 0.487$

Marginal: $P(L = \text{Bad}) = 0.2045 + 0.2234 + 0.0255 + 0.0442 = 0.498$

Joint: $P(\text{High}, \text{Hard}|\text{Bad}) = 0.0442/0.498 = 0.089$

Each parent: $P(\text{High}|\text{Bad}) = 0.0697/0.498 = 0.140$

$P(\text{Hard}|\text{Bad}) = 0.2676/0.498 = 0.538$

Product: $0.140 \cdot 0.538 = 0.075 \neq 0.089$

- As S2.08, with $p(C|i, d)$ replaced by a sum over the unobserved grade
- $w(i, d)$ depends on i and d together, so the rows do not factor
  - the parents are coupled

3. Three-node patterns

S3.09 · Slide 30/56

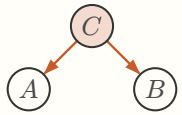
STUDY NOTE

The calculation is the one for a grade of C, except that the probability of a C is replaced by a sum over the unobserved grade, weighted by the probability of a bad letter. Because these weights depend on intelligence and difficulty together, the rows do not factor into a function of each, and the two parents are coupled.

YOUR NOTES

Lecture L03 — Graphical Models

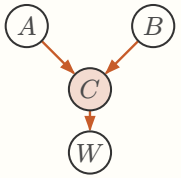
Section 3 summary: Forks and pipes block when observed; colliders open



fork: observing C blocks



pipe: observing C blocks



collider: observing C or W opens

TAKE HOME

- Observing the middle of a fork or pipe blocks it.
- Observing a collider, or a descendant of it, opens it.

STUDY NOTE

Three arrangements of three variables decide every independence in a directed graph. In a fork and in a pipe, observing the middle variable blocks the dependence between the ends. In a collider the ends are independent until the middle variable or one of its descendants is observed, which makes them dependent.

YOUR NOTES

Section 4

Whole graphs: paths, guarantees and blankets

In a large graph, when is one variable independent of another, given what we know?

YOUR NOTES

Lecture L03 — Graphical Models

An open path closed by information: SAT and letter

Path $S \leftarrow I \rightarrow G \rightarrow L$: no edge joins the SAT and the letter

Two nodes inside the path: I, a fork; G, a pipe.

Nothing observed: both pass, the path is open

$P(S = \text{Good}, L = \text{Good}) = 0.198$

$P(S = \text{Good}) \cdot P(L = \text{Good}) = 0.275 \cdot 0.502 = 0.138$

A good SAT score suggests intelligence, a good grade, a good letter **dependent**

Grade observed, $G = A$: the pipe blocks, the path is closed

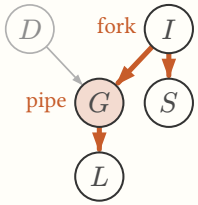
$P(S = \text{Good}, L = \text{Good} | A) = 0.459$

$P(S = \text{Good} | A) \cdot P(L = \text{Good} | A) = 0.510 \cdot 0.90 = 0.459$

Once the grade is known, the letter has nothing more to learn from the SAT

independent

- One observed fork or pipe node is enough to close a path



Murphy B2 Fig. 4.2

STUDY NOTE

The SAT and the letter are joined by a path through intelligence and the grade. With nothing observed, both interior nodes let dependence through, so a good SAT raises the chance of a good letter from 0.50 to 0.72. Observing the grade closes the path, as for the pipe, and the two become independent.

YOUR NOTES

Lecture L03 — Graphical Models

A closed path opened by information: SAT and difficulty

Path $S \leftarrow I \rightarrow G \leftarrow D$: no edge joins the SAT and the difficulty

Two nodes inside the path: I, a fork; G, a collider.

Nothing observed: the collider blocks, the path is closed

$P(S = \text{Good}, D = \text{Hard}) = 0.110 = P(S = \text{Good}) \cdot P(D = \text{Hard}) = 0.275 \cdot 0.40$

The SAT says nothing about the class's difficulty **independent**

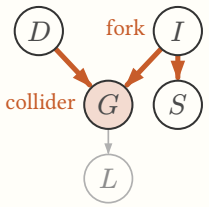
Grade observed, $G = C$: the collider passes, the path is open

$P(S = \text{Good}, D = \text{Hard} | C) = 0.083$

$P(S = \text{Good} | C) \cdot P(D = \text{Hard} | C) = 0.109 \cdot 0.629 = 0.069$

A good SAT makes a hard class likelier: $0.63 \rightarrow 0.76$ **dependent**

- Observing a collider, or a descendant of it, opens a path that was closed



Murphy B2 Fig. 4.2

STUDY NOTE

The SAT and difficulty are joined by a path through intelligence and the grade, but the grade is a collider on this path, so with nothing observed the path is closed and they are independent. Observing a C opens the collider, and a good SAT then makes a hard class more likely. Murphy reports the same probabilities.

YOUR NOTES

Lecture L03 — Graphical Models

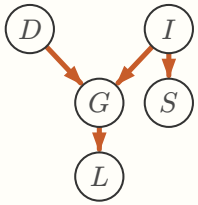
A path is open only if every node along it lets dependence through

From the two examples to a rule for any path

- A *path*: a sequence of distinct nodes, consecutive ones joined by an edge in either direction.
- A node inside it is a *collider on the path* if both path edges point into it.

given evidence E, an interior node blocks if		from
it is a fork or pipe node	it is in E	S3.03, S3.05
it is a collider	neither it nor a descendant is in E	S3.07–S3.08

A path is *open* (active) if no interior node blocks it.



Murphy B2 Fig. 4.2

- A long path is closed as soon as one node along it blocks

STUDY NOTE

The two examples combine into one rule for any path. Given what is observed, a fork or pipe node blocks the path if it is observed, and a collider blocks it unless it or one of its descendants is observed. A path is open only if none of its interior nodes blocks it.

YOUR NOTES

Lecture L03 — Graphical Models

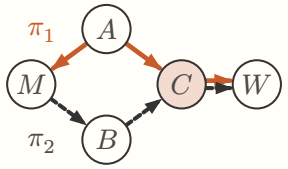
d-separation: independent once every path between them is closed

Two paths from M to W: is closing one enough?

Path π_1 runs through A, path π_2 through B; both end $C \to W$.

X and Y are *d-separated* by E if **every** path between them is closed. Then $X \perp Y \mid E$ for every set of tables.

observed	π_1 (A fork, C pipe)	π_2 (B pipe, C pipe)
A	closed at A	open — no guarantee
A, B	closed at A	closed at B: $M \perp W \mid \{A, B\}$
C	closed at C	closed at C: $M \perp W \mid C$



Course-constructed

Not proved here

Book 2, §4.2.4.1 defines d-separation and §4.2.4.4 states this result; its proof is in Koller and Friedman (2009), *Probabilistic Graphical Models*.

- Every path must be closed somewhere, so observing A leaves π_2 open until B is observed
- One node can close several paths, as C, a pipe on both, does

4. Paths and blankets

S4.05 · Slide 36/56

STUDY NOTE

Two variables can be joined by several paths, and closing one is not enough. If every path between them is closed by the observed set, they are d-separated, and then they are independent for every choice of tables. One node can close several paths at once, as C does here. The result is stated, not proved, in this course.

YOUR NOTES

Lecture L03 — Graphical Models

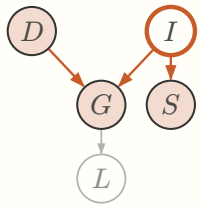
A Markov blanket: given it, nothing else tells us more about a variable

Markov blanket: $p(x_i \mid x_{-i}) = p\left(x_i \mid x_{\mathrm{mb}(i)}\right)$

In a DAG: $\mathrm{mb}(i) = \mathrm{pa}(i) \cup \mathrm{ch}(i) \cup \mathrm{cop}(i)$

Student record: $\mathrm{mb}(I) = \varnothing \cup \{G, S\} \cup \{D\}$

$p(i \mid d, g, s, l) = p(i \mid d, g, s) \propto p(i)\, p(g|i, d)\, p(s|i)$



Murphy B2 Fig. 4.2

- x_{-i} is every variable except x_i
- Given its blanket, the rest tells us nothing more about x_i
- Parents $\mathrm{pa}(i)$, children $\mathrm{ch}(i)$, and co-parents $\mathrm{cop}(i)$: the children's other parents
- Parents and children close the paths through them
  - a child is a collider for its other parents, so they join (S4.04)
- Only the factors containing i remain, so the letter drops out

4. Paths and blankets

S4.06 · Slide 37/56

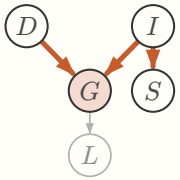
STUDY NOTE

A variable's Markov blanket is a set of variables that, once known, makes the rest irrelevant to it. In a directed graph it consists of the parents, the children and the children's other parents. The children close the paths through them, but observing a child opens the path to its other parents, so they are included too.

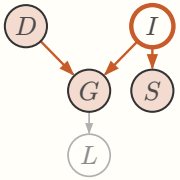
YOUR NOTES

Lecture L03 — Graphical Models

Section 4 summary: Paths show what the graph guarantees



observing the collider G opens $S - I - G - D$



I's Markov blanket: G, S, D

TAKE HOME

- Observed forks and pipes close paths; observed colliders open them.
- Every path closed: independent.
- A Markov blanket closes all paths to the rest.

STUDY NOTE

Section 4 turned the three-node patterns into a rule for whole graphs. A path is closed by an observed fork or pipe node, or by an unobserved collider, and variables are independent when every path between them is closed. A Markov blanket closes every path from one variable to the rest.

YOUR NOTES

Section 5

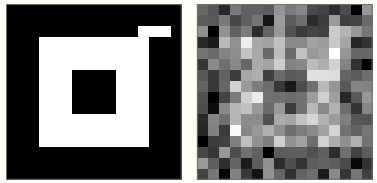
Undirected models and symmetric potentials

Potentials, a global normalizer and graph separation

YOUR NOTES

Lecture L03 — Graphical Models

Symmetric relationships: restore a binary image



clean z observed y

Course-generated image

Want: $z = (z_1, \ldots, z_{256}), \quad z_i \in \{-1, +1\}$

See: $y_i = z_i + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2), \quad \sigma^2 = 0.8$

Notation, following Murphy (Book 2 §4.3)

$i \sim j$	pixels i and j are neighbours (each pair counted once)
$\mathrm{ne}(i)$	the neighbours of pixel i: four, fewer at the edge
z_{-i}, $z_{\mathrm{ne}(i)}$	all pixels except i; its neighbours

- -1 is dark, $+1$ light
- The clean image is used only to score a restoration
- The noise is independent across pixels
- To judge a guess z: how likely is y under it, and how plausible is it as an image?

5. Undirected models

S5.02 · Slide 40/56

STUDY NOTE

The second half of the lecture restores a black-and-white image from noisy measurements. Each pixel is dark or light, and each measurement is its pixel's value plus independent Gaussian noise. To judge a candidate image we ask how well it explains the measurements and how plausible it is as an image. The clean image is used only for scoring.

YOUR NOTES

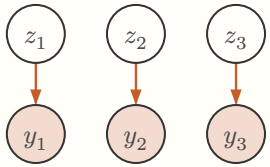
Lecture L03 — Graphical Models

A guessed clean image gives a likelihood for the measurements

One pixel: $p(y_i \mid z_i) = \mathcal{N}(y_i \mid z_i, \sigma^2)$

All pixels: $p(\boldsymbol{y} \mid \boldsymbol{z}) = \prod_i \mathcal{N}(y_i \mid z_i, \sigma^2)$

$y_i = 0.3$	$z_i = +1$	$z_i = -1$
$\mathcal{N}(0.3 \mid z_i, 0.8)$	0.328	0.155



Likelihood-only estimate: $\hat{z}_i = \arg\max_{z_i} \mathcal{N}(y_i \mid z_i, \sigma^2) = \operatorname{sign}(y_i)$



$\operatorname{sign}(\boldsymbol{y})$: 42 errors of 256

Course-generated image

- Each measurement depends only on its own pixel
- Independent noise gives a product over pixels, and one σ^2 for all is parameter tying (S1.09)
- A reading of 0.3 favours $+1$, by a factor of 2.1
- The lecture's only estimate, from the likelihood alone, is the pixel-by-pixel guess

STUDY NOTE

Each measurement depends only on its own pixel, so the likelihood of a candidate image is a product over pixels, with one noise variance for all of them. Maximizing the likelihood alone gives the pixel-by-pixel guess, which gets 42 of 256 pixels wrong. It is the only estimate this lecture makes; the notebook chooses estimates from the posterior.

YOUR NOTES

Lecture L03 — Graphical Models

A prior over whole images: given its neighbours, a pixel ignores the rest

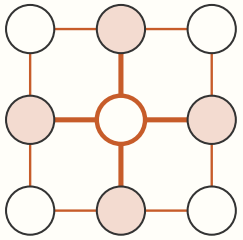
Carried forward S1.04 · Conditional independence
$X \perp Y \mid U \quad \Leftrightarrow \quad p(x|y,u) = p(x|u)$

Prior: $p(\boldsymbol{z}) = p(z_1, z_2, \ldots, z_{256}), \quad \boldsymbol{z} \in \{-1, +1\}^{256}$

Rest of the image: $\boldsymbol{z}_{-i} = (z_1, \ldots, z_{i-1}, z_{i+1}, \ldots, z_{256})$

Neighbours: $\boldsymbol{z}_{\mathrm{ne}(i)} = \big(z_j : j \in \mathrm{ne}(i)\big), \quad \mathrm{ne}(i) = \text{up, down, left, right}$

Neighbours suffice: $p(z_i \mid \boldsymbol{z}_{-i}) = p\Big(z_i \mid \boldsymbol{z}_{\mathrm{ne}(i)}\Big)$



- 2^{256} images are far too many to tabulate, as for S1's full model of a name
- $\boldsymbol{z}_{-i}$ is every pixel except i
- $\boldsymbol{z}_{\mathrm{ne}(i)}$ is only the four pixels around it (shaded)
- Images have regions, so the four neighbours tell us all the rest can about pixel i
  - a conditional independence, as in L02's names model
  - around the pixel rather than before it

5. Undirected models

S5.04 · Slide 42/56

STUDY NOTE

A prior over whole images must assign a probability to each of 2 to the power 256 images, far too many to tabulate. Real images have regions, which suggests a conditional-independence assumption: given its four neighbours, a pixel's value does not depend on the rest of the image. It plays the role L02's assumption played for names.

YOUR NOTES

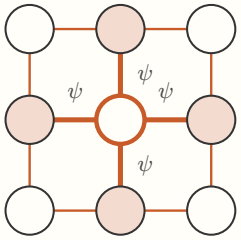
Lecture L03 — Graphical Models

Without a natural pixel order, build the prior from pair weights, normalized once

Carried forward S2.06 · A DAG's factors
$p(\boldsymbol{x}) = \prod_i p\big(x_i \mid \boldsymbol{x}_{\mathrm{pa}(i)}\big)$: one table per variable, given its parents

Pair weights: $p(\boldsymbol{z}) = \frac{1}{Z} \prod_{i \sim j} \psi_{ij}(z_i, z_j), \quad \psi_{ij} \geq 0$

Normalizer: $Z = \sum_{\boldsymbol{z}} \prod_{i \sim j} \psi_{ij}(z_i, z_j)$



- A DAG's factors come from the chain rule in one order of the variables
  - each table conditions on parents that come earlier in that order
- In any pixel order, some factors keep more than the neighbours (Murphy, Book 2 §4.3.1)
  - a pixel has neighbours on every side, so some come later in the order
- Instead, give each pair of neighbours one nonnegative weight
  - a pixel enters only through its pairs
- The weights need not sum to one, so a single Z normalizes the product
  - this form implies S5.04's assumption (S5.11)

5. Undirected models

S5.05 · Slide 43/56

STUDY NOTE

A directed model needs an order of the variables, but in every order of the pixels some factors would keep more than the neighbours. Undirected models need no order: they give each pair of neighbours a nonnegative weight and divide the product by one normalizing constant, Z. A distribution of this form satisfies the neighbour assumption.

YOUR NOTES

Lecture L03 — Graphical Models

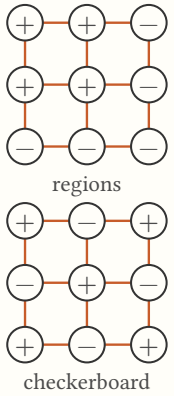
Choosing the pair weight: more weight when neighbours agree

Carried forward S5.05 · Pair weights
$p(\boldsymbol{z}) = \frac{1}{Z}\prod_{i\sim j}\psi_{ij}(z_i, z_j)$

$$\psi_{ij}(z_i, z_j) = e^{J z_i z_j}$$

	weight	$J = 0.5$
$z_i = z_j$	e^{J}	1.65
$z_i \neq z_j$	e^{-J}	0.61

Regions: $e^{0.5(8-4)} = e^{2} \approx 7.4$
Checkerboard: $e^{0.5(0-12)} = e^{-6} \approx 0.0025$



- One weight shared by all pairs, set by a coupling J (parameter tying, as with P in S1.09)
- $J > 0$ pulls neighbours to agree; $J = 0$ weighs all pairs equally
- $z_i z_j = \pm 1$, so the exponent is $J \times$ (agreeing − disagreeing pairs), here 8 − 4 and 0 − 12
- Z sums $2^9 = 512$ patches here, 2^{256} images for the scan

5. Undirected models

S5.06 · Slide 44/56

STUDY NOTE

The Ising model gives every neighbouring pair the same weight function, larger when the two pixels agree, with its strength set by the coupling J. Using one coupling for all pairs is parameter tying again. With J = 0.5, a patch made of two regions gets weight 7.4 and a checkerboard 0.0025, before dividing by Z.

YOUR NOTES

Lecture L03 — Graphical Models

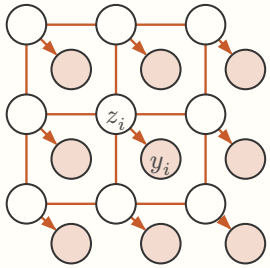
Prior times likelihood: one model for the clean image and its measurements

Carried forward S5.03, S5.05 · Likelihood and prior
$p(\boldsymbol{y} \mid \boldsymbol{z}) = \prod_i \mathcal{N}(y_i \mid z_i, \sigma^2)$, $p(\boldsymbol{z}) = \frac{1}{Z} \prod_{i \sim j} \psi_{ij}(z_i, z_j)$

Joint: $p(\boldsymbol{y}, \boldsymbol{z}) = \frac{1}{Z} \prod_{i \sim j} \psi_{ij}(z_i, z_j) \prod_i \mathcal{N}(y_i \mid z_i, \sigma^2)$

Posterior: $p(\boldsymbol{z} \mid \boldsymbol{y}) = \frac{1}{Z\, p(\boldsymbol{y})} \prod_{i \sim j} \psi_{ij}(z_i, z_j) \prod_i \varphi_i(z_i)$

Data weight: $\varphi_i(z_i) = \mathcal{N}(y_i \mid z_i, \sigma^2)$



- Murphy's restoration model (Book 2 Eq. 4.117): undirected among pixels, directed to measurements
- With $\boldsymbol{y}$ fixed, each φ_i is a weight on one pixel
  - $y_i = 0.3$ weighs $+1$ by 0.328, -1 by 0.155
- The lecture computes this posterior
  - its normalizer $Z\, p(\boldsymbol{y})$ also sums over all images
  - choosing a restored image from it is the notebook's task

STUDY NOTE

Multiplying the prior by the likelihood gives Murphy's restoration model; his Equation 4.117 prints a sum where the factors multiply. With the measurements fixed, each likelihood factor is a weight on one pixel, and the posterior is the product of the pair weights and pixel weights divided by its normalizer. This lecture computes the posterior.

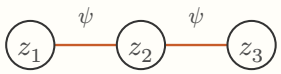
YOUR NOTES

Lecture L03 — Graphical Models

A three-pixel image can be solved by brute force

Each image's weight: one pair weight per neighbouring pair, $J = 0.5$

$z_1 z_2 z_3$	pairs that agree	weight	images	probability
$+++,\ ---$	both	$e^1 = 2.72$	2	0.267 each
$++-,\ -++,\ --+,\ +--$	one	$e^0 = 1$	4	0.098 each
$+-+,\ -+-$	neither	$e^{-1} = 0.37$	2	0.036 each



Z: the total weight of all eight images

$Z = 2e + 4 + 2e^{-1} \approx 10.17$

- Eight images can be listed
- The 16×16 scan has 2^{256}, so its Z cannot

STUDY NOTE

For a row of three pixels every image can be listed. Each image's weight is the product of its two pair weights, Z is the total weight of all eight images, and each image's probability is its weight divided by Z. For the 256-pixel image the same list would have 2 to the power 256 entries.

YOUR NOTES

Lecture L03 — Graphical Models

The coupling J decides when neighbours outweigh a pixel's own reading

All eight images for $y = (0.9, -0.3, 0.9)$, $J = 0.5$

$z_1 z_2 z_3$	prior weight	data weight	product	posterior
$+++$	2.72	0.0305	0.0828	0.644
$++-$	1	0.0032	0.0032	0.025
$+-+$	0.37	0.0645	0.0237	0.184
$+--$	1	0.0068	0.0068	0.053
$-++$	1	0.0032	0.0032	0.025
$-+-$	0.37	0.0003	0.0001	0.001
$--+$	1	0.0068	0.0068	0.053
$---$	2.72	0.0007	0.0019	0.015

$P(z_2 = +1 \mid y)$, the middle pixel's posterior

y_2	$J = 0$	$J = 0.5$	$J = 1$
-0.3	0.32	0.69	0.89
-1.0	0.08	0.28	0.59

- Posterior = weight $\psi_{12}\,\psi_{23}\,\varphi_1\,\varphi_2\,\varphi_3$ (S5.07) ÷ total, as in S2.08
- At $J = 0$ the posterior follows the reading alone
- At $J = 0.5$ the neighbours outweigh a weak dark reading
- A strong dark reading outweighs $J = 0.5$ but not $J = 1$
  - a large J removes noise but can erase a real thin line

STUDY NOTE

Multiplying each image's prior weight by its data weights and dividing by the total gives the posterior. With no coupling the middle pixel follows its own reading. With J = 0.5, clearly light neighbours outweigh a weak dark reading but not a strong one; with J = 1 they outweigh both, which can erase a real thin line.

YOUR NOTES

Lecture L03 — Graphical Models

In general: a nonnegative potential on each clique, and one normalizer

Carried forward S5.05 · Pair weights
$p(\boldsymbol{z}) = \frac{1}{Z}\prod_{i\sim j}\psi_{ij}(z_i, z_j)$



Undirected model, defined by: $p(\boldsymbol{x}) = \frac{1}{Z}\prod_c \psi_c(\boldsymbol{x}_c)$
Normalizer: $Z = \sum_{\boldsymbol{x}}\prod_c \psi_c(\boldsymbol{x}_c)$

- c runs over *cliques*, sets of mutually adjacent nodes
  - in the image prior, neighbouring pairs
  - the posterior adds single pixels, the data weights φ_i
- *Potentials* $\psi_c \geq 0$ are not CPDs: rows need not sum to one
- Z plays the role of L02's $Z(\eta)$: the sum of unnormalized weights

5. Undirected models

S5.10 · Slide 48/56

STUDY NOTE

In general an undirected model is a product of nonnegative potentials, one for each clique of mutually adjacent nodes, divided by a single normalizer Z. Potentials are not conditional distributions, and their rows need not sum to one. Z plays the same role as the normalizer of an exponential family in L02.

YOUR NOTES

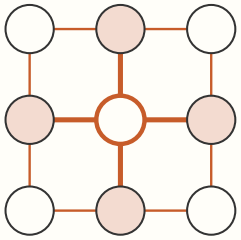
Lecture L03 — Graphical Models

A pixel's neighbours are its Markov blanket

Carried forward S4.05 · d-separation
every path between X and Y closed by $E \quad \Rightarrow \quad X \perp Y \mid E$

Separation: E separates X from $Y \quad \Rightarrow \quad X \perp Y \mid E$

Markov blanket: $p(z_i \mid \boldsymbol{z}_{-i}) = p\left(z_i \mid \boldsymbol{z}_{\mathrm{ne}(i)}\right)$



- E separates X from Y if every path between them passes through E
  - then $X \perp Y \mid E$ for every choice of potentials
  - on the row, z_2 separates z_1 from z_3
- Simpler than d-separation: no colliders, so any observed node on a path closes it
- A pixel's neighbours close every path out of it
  - so the product form makes S5.04's assumption
- Stated, not proved here: Murphy, Book 2 §4.3.6.1

5. Undirected models

S5.11 · Slide 49/56

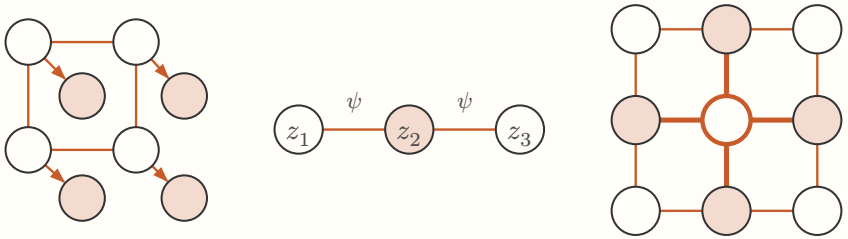
STUDY NOTE

In an undirected graph, an observed set separates two nodes if every path between them passes through it, and then the two are independent for every choice of potentials. There are no colliders, so the rule is simpler than d-separation. A pixel's four neighbours separate it from the rest of the image: they are its Markov blanket.

YOUR NOTES

Lecture L03 — Graphical Models

Section 5 summary: Pair weights and data weights, one normalizer



TAKE HOME

- Likelihood × prior: pair weights and one-pixel weights; normalizers sum over all images.
- Observed nodes separate.
- In the prior, a pixel's blanket is its neighbours.

STUDY NOTE

Section 5 modelled image restoration with an undirected graph. The likelihood gives each pixel a weight from its measurement, the prior gives each neighbouring pair a weight, and normalizers turn the products into distributions. Observed nodes separate whatever they cut off, so in the prior each pixel's Markov blanket is its neighbours.

YOUR NOTES

Section 6

The Ising prior: a local site conditional and a global normalizer

What a coupling says about one pixel, and which questions stay global

YOUR NOTES

Lecture L03 — Graphical Models

The conditional distribution $p(z_i \mid \boldsymbol{z}_{-i})$ does not depend on Z

Carried forward S5.06, S5.11

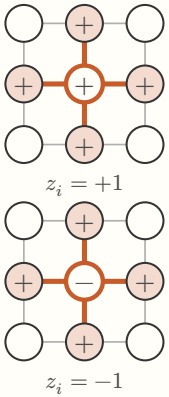
$\psi_{ij}(z_i, z_j) = e^{J z_i z_j}$; $p(z_i \mid \boldsymbol{z}_{-i}) = p\big(z_i \mid \boldsymbol{z}_{\mathrm{ne}(i)}\big)$

Weight: $\mathrm{weight}(\boldsymbol{z}) = \prod_{j \in \mathrm{ne}(i)} e^{J z_i z_j} \cdot R, \quad R = \prod_{k \sim l,\, i \notin \{k,l\}} e^{J z_k z_l}$

Neighbours: $\prod_{j \in \mathrm{ne}(i)} e^{J z_i z_j} = e^{z_i h_i}, \quad h_i \triangleq J \sum_{j \in \mathrm{ne}(i)} z_j$

$$P(z_i = +1 | \boldsymbol{z}_{-i}) = \frac{p(+1, \boldsymbol{z}_{-i})}{\sum_{s=\pm 1} p(s, \boldsymbol{z}_{-i})}$$

$$= \frac{\cancel{Z^{-1}} R\, e^{h_i}}{\cancel{Z^{-1}} R (e^{h_i} + e^{-h_i})} = \frac{1}{1 + e^{-2h_i}}$$



- With every other pixel fixed, Z and the pairs not touching i are constants
- R is the product of the pair weights not touching i, so it contains no z_i
- h_i adds up the neighbours, weighted by J
- Z and R cancel, leaving log-odds $2h_i$ from four neighbours, whatever the image size

6. Ising prior

S6.02 · Slide 52/56

STUDY NOTE

Fix every pixel except one and compare its two values. The normalizer Z and every pair weight not involving that pixel are the same in both cases and cancel. What remains depends only on the four neighbours through the sum h: the probability that the pixel is light is one over one plus e to the power minus 2h.

YOUR NOTES

Lecture L03 — Graphical Models

Four agreeing neighbours give 0.982: a Bernoulli with log-odds $2h_i$

Carried forward S6.02 · A pixel given the rest

$P(z_i = +1|\boldsymbol{z}_{-i}) = 1/\left(1 + e^{-2h_i}\right), \quad h_i = J \sum_{j \in \text{ne}(i)} z_j$

$J = 0.5$

neighbours	$\sum_j z_j$	h_i	$P(z_i = +1\|\boldsymbol{z}_{-i}, J)$
all four $+1$	4	2	0.982
three $+1$, one -1	2	1	0.881
two and two	0	0	1/2

Carried over L02 S1.04 · Bernoulli in natural coordinates

L02: $P(x = 1) = \theta(\eta) = 1/(1 + e^{-\eta})$, η the log odds

Here: $x = 1 \Leftrightarrow z_i = +1$, $\eta = 2h_i \;\Rightarrow\; P(z_i = +1|\boldsymbol{z}_{-i}) = 1/\left(1 + e^{-2h_i}\right)$

- Given the rest, a pixel is L02's Bernoulli with log odds $2h_i$
  - each $+1$ neighbour adds $2J$ to the log odds
  - each -1 neighbour subtracts $2J$

6. Ising prior

S6.03 · Slide 53/56

STUDY NOTE

With J = 0.5, four light neighbours make a light pixel 98 percent likely, three make it 88 percent likely, and a two-two split leaves it at one half. The formula is L02's Bernoulli distribution with log odds 2h: each light neighbour adds 2J to the log odds and each dark neighbour subtracts 2J.

YOUR NOTES

Lecture L03 — Graphical Models

A conditional is not a marginal: Z and posterior marginals are global

question	cost	why
$P(z_i = +1 \mid \boldsymbol{z}_{-i})$	four neighbours	Z and other pairs cancel (S6.02)
$P(z_i = +1) = 1/2$	free	symmetry: flipping every pixel keeps every weight
$Z(J)$	2^{256} terms	a sum over every image
$P(z_i = +1 \mid \boldsymbol{y})$	2^{255} terms	a sum over every other pixel, since the data break the symmetry

- A pixel's conditional is cheap
- It is not its marginal
  - 0.982 given four agreeing neighbours, 1/2 with nothing known
- Z needs every image
- With data, a pixel's posterior marginal sums over all the other pixels

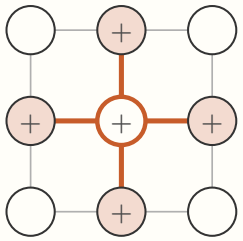
STUDY NOTE

A pixel's probability given all the other pixels needs only its four neighbours. Its probability given nothing is one half, by symmetry, so the conditional and the marginal differ. Computing Z requires a sum over every image, and once data are included, a pixel's posterior probability requires a sum over all the other pixels.

YOUR NOTES

Lecture L03 — Graphical Models

Section 6 summary: A local site conditional, a global normalizer



four agreeing neighbours: $P(z_i = +1 | \boldsymbol{z}_{-i}) = 0.982$
no neighbour known: $P(z_i = +1) = 1/2$

TAKE HOME

- A pixel's conditional is cheap: four neighbours, no Z.
- A conditional is not a marginal.
- Z and posterior marginals are global.

STUDY NOTE

Under the Ising prior, each pixel's distribution given all the others is a Bernoulli whose log odds add up its neighbours' values; Z cancels. That conditional is not the pixel's marginal probability, and Z and the posterior marginals still require sums over all images.

YOUR NOTES

Lecture L03 — Graphical Models

Notebook: restore a noisy image, and find out whether coupling helps

What the notebook asks

The lecture built the posterior and showed what is cheap in it: one pixel given all the others needs only its neighbours, while each pixel's posterior marginal, like Z, needs every other pixel. Restoring a scan means choosing one image from the posterior. The notebook makes that choice with a cheap local method and tests whether the coupling J reduces errors.

What you will do

1. **Add the data to S6.02's conditional.** Each reading adds y_i/σ^2 to the field: $h_i = J\sum_{j\in\,\mathrm{ne}(i)} z_j + y_i/\sigma^2$.
2. **Restore the image one pixel at a time.** Iterated conditional modes (ICM) sets each pixel in turn to its conditional's more probable value, $z_i \leftarrow \mathrm{sign}(h_i)$, until a pass changes nothing. No step lowers the posterior.
3. **Compare with sign(y).** Count pixel errors with and without coupling; choose J on a separate calibration image.
4. **Ask what ICM computed.** An image no single flip improves, not necessarily the most probable one, and no probabilities. Then criticize the prior: it erodes lines one pixel wide.

You will use S5.03–S5.07 (the model), S6.02–S6.03 (the conditional) and L01's Bayes' rule and Gaussian density. The clean image only scores a restoration.

STUDY NOTE

The notebook adds the data to Section 6's conditional and uses it to restore a larger noisy image one pixel at a time. It asks whether coupling neighbours reduces errors compared with reading each pixel alone, how to choose J without using the answer, and what the resulting image is and is not guaranteed to be.

YOUR NOTES